

Configuration	A	B	C	D	E	F	G	H	All constraints	Argument
Node 1.1.1.6 in Figure 4b	0	1	1	1	0	1	1	1	1	0
Node 2.1-9 in Figure 4d	0	0	0	1	0	0	0	0	1	1

No coding, no tools—just direct deduction to solve the following problem.

Logical Argument

Variables:

{A, B, C, D, E, F, G, H}

Constraints:

- (1) $(A \oplus B) \rightarrow C$
- (2) $(C \wedge D) \rightarrow \neg E$
- (3) $(E \vee F) \leftrightarrow G$
- (4) $(\neg A \wedge B) \rightarrow (D \wedge F)$
- (5) $(G \wedge H) \rightarrow (A \vee B)$
- (6) $(A \wedge \neg B) \rightarrow (C \wedge \neg E)$
- (7) $(F \oplus H) \rightarrow (D \vee E)$
- (8) $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$
- (9) $(D \wedge E) \rightarrow (\neg G \vee F)$
- (10) $(H \vee G) \rightarrow (A \oplus B)$
- (11) $(\neg E \wedge F) \leftrightarrow (C \wedge D)$
- (12) $(A \vee C) \rightarrow (B \vee \neg F)$
- (13) $(G \wedge \neg H) \rightarrow (E \vee \neg A)$
- (14) $(B \wedge F) \rightarrow (\neg D \vee H)$
- (15) $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$

Step by step deduct all configurations that satisfy all constraints and validate the argument:

$$(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$$

Four travelers language deduction puzzle

No coding, no tools—just direct deduction to solve the following problem.

Logical Argument
Variables:
{A, B, C, D, E, F, G, H}
Constraints:
(1) $(A \oplus B) \rightarrow C$
(2) $(C \wedge D) \rightarrow \neg E$
(3) $(E \vee F) \leftrightarrow G$
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(2) $(C \wedge D) \rightarrow \neg E$
(3) $(E \vee F) \leftrightarrow G$
(4) $(\neg A \wedge B) \rightarrow (D \wedge F)$
(5) $(G \wedge H) \rightarrow (A \vee B)$
(6) $(A \wedge \neg B) \rightarrow (C \wedge \neg E)$
(7) $(F \oplus H) \rightarrow (D \vee E)$
(8) $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$
(9) $(D \wedge E) \rightarrow (\neg G \vee F)$
(10) $(H \vee G) \rightarrow (A \oplus B)$
(11) $(\neg E \wedge F) \leftrightarrow (C \wedge D)$
(12) $(A \vee C) \rightarrow (B \vee \neg F)$
(13) $(G \wedge \neg H) \rightarrow (E \vee \neg A)$

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(13) $(G \wedge \neg H) \rightarrow (E \vee \neg A)$
 (14) $(B \wedge F) \rightarrow (\neg D \vee H)$
 (15) $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$
 Step by step deduct all configurations that satisfy all constraints and validate the argument: $(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$
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I'll work through this systematically to find all valid configurations and then validate the argument.

Step 1: Analyze key constraints to narrow possibilities

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Step 1: Analyze key constraints to narrow possibilities

Let me start by examining some critical constraint relationships.

From constraint (11): $(\neg E \wedge F) \leftrightarrow (C \wedge D)$
 This means: $\neg E \wedge F$ is true iff $C \wedge D$ is true

From constraint (2): $(C \wedge D) \rightarrow \neg E$
 This means: if $C \wedge D$ is true, then E is false

Combining these: If $C \wedge D$ is true, then $\neg E$ is true. For (11) to hold, we need F to be true.
 So: $C \wedge D \rightarrow (\neg E \wedge F)$

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SO: $C \wedge D \rightarrow (\neg E \wedge F)$

Also from (11): If $\neg E \wedge F$ is true, then $C \wedge D$ must be true.

Step 2: Analyze constraint (3)

$(E \vee F) \leftrightarrow G$ means:

- G is true iff at least one of E or F is true
- G is false iff both E and F are false

Step 3: Consider cases based on $C \wedge D$

Case 1: $C \wedge D = \text{True}$ From (2): $E = \text{False}$ From (11): $\neg E \wedge F$ must be true, so $F = \text{True}$ From (3):

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Case 1: $C \wedge D = \text{True}$ From (2): $E = \text{False}$ From (11): $\neg E \wedge F$ must be true, so $F = \text{True}$ From (3):
Since $F = \text{True}$, $G = \text{True}$

Now with $C=T, D=T, E=F, F=T, G=T$:

From (10): $(H \vee G) \rightarrow (A \oplus B)$

Since $G=T$, we have $T \rightarrow (A \oplus B)$, so $A \oplus B$ must be true.

From (1): $(A \oplus B) \rightarrow C$

Since $A \oplus B=T$ and $C=T$, this is satisfied.

Subcase 1a: $A=T, B=F$ (so $A \oplus B=T$) From (6). $\neg B \rightarrow (C \wedge \neg E)$ We have $T \wedge T \rightarrow T \wedge T$,
which gives $T \rightarrow T$.

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Subcase 1a: $A=T, B=F$ (so $A \oplus B=T$) From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E)$ We have $T \wedge T \rightarrow T \wedge T$, which gives $T \rightarrow T \checkmark$

From (4): $(\neg A \wedge B) \rightarrow (D \wedge F)$
We have $F \wedge F \rightarrow T \wedge T$, which gives $F \rightarrow T \checkmark$

From (12): $(A \vee C) \rightarrow (B \vee \neg F)$
We have $T \rightarrow (F \vee F) = T \rightarrow F \times$

So $A=T, B=F$ is invalid in Case 1.

Subcase 1b: $A=F, B=T$ (so $A \oplus B=T$) From (4): $(\neg A \wedge B) \rightarrow (D \wedge F)$ We have $T \wedge T \rightarrow T \wedge T$, which gives $T \rightarrow T \checkmark$

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So $A=T, B=F$ is invalid in Case 1.

Subcase 1b: $A=F, B=T$ (so $A \oplus B=T$) From (4): $(\neg A \wedge B) \rightarrow (D \wedge F)$ We have $T \wedge T \rightarrow T \wedge T$, which gives $T \rightarrow T \checkmark$

From (12): $(A \vee C) \rightarrow (B \vee \neg F)$
We have $T \rightarrow (T \vee F) = T \rightarrow T \checkmark$

Now for H, let's test both values:

H = True: From (5): $(G \wedge H) \rightarrow (A \vee B) = T \rightarrow T \checkmark$ From (7): $(F \oplus H) \rightarrow (D \vee E) = F \rightarrow T \checkmark$ From (13): $(G \wedge \neg H) \rightarrow (E \vee \neg A) = F \rightarrow T \checkmark$ From (14): $(\neg F) \rightarrow (\neg D \vee H) = T \rightarrow T \checkmark$ From (15): $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$ LHS: $T \vee T = T$ RHS: $(F \oplus T) \vee (T \wedge F) = T \vee F = T \checkmark$ From (8): $(\neg A \wedge$

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$$H \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$$
 LHS: $T \vee T = T$ RHS: $(F \oplus T) \vee (T \wedge F) = T \vee F = T \checkmark$ From (8): $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$ LHS: $T \wedge T = T$ RHS: $T \vee F = T \checkmark$ From (9): $(D \wedge E) \rightarrow (\neg G \vee F) = F \rightarrow T \checkmark$

Configuration 1: A=F, B=T, C=T, D=T, E=F, F=T, G=T, H=T ✓

H = False: From (5): $(G \wedge H) \rightarrow (A \vee B) = F \rightarrow T \checkmark$ From (7): $(F \oplus H) \rightarrow (D \vee E) = T \rightarrow T \checkmark$ From (13): $(G \wedge \neg H) \rightarrow (E \vee \neg A) = T \rightarrow T \checkmark$ From (14): $(B \wedge F) \rightarrow (\neg D \vee H) = T \rightarrow F \times$

So $H=F$ is invalid in Case 1.

Case 2: C ∧ D = False

From (11): $\neg E \wedge F$ must be false

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From (11): $\neg E \wedge F$ must be false
This means: $E \vee \neg F$ is true

Subcase 2a: C=F From (3): Need to determine G based on E and F

From (10): $(H \vee G) \rightarrow (A \oplus B)$
If $H \vee G$ is true, then $A \oplus B$ must be true.

From (1): $(A \oplus B) \rightarrow C = (A \oplus B) \rightarrow F$
If $A \oplus B$ is true, then C must be true, but $C=F$, so $A \oplus B$ must be false.
Therefore $A \oplus B = F$, meaning $A=B$.

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From (1): $(A \oplus B) \rightarrow C = (A \oplus B) \rightarrow F$
 If $A \oplus B$ is true, then C must be true, but $C=F$, so $A \oplus B$ must be false.
 Therefore $A \oplus B = F$, meaning $A=B$.

If $A \oplus B = F$, then from (10), $H \vee G$ must be false, so $H=F$ and $G=F$.

From (3): $(E \vee F) \leftrightarrow G = (E \vee F) \leftrightarrow F$
 This means $E \vee F = F$, so $E=F$ and $F=F$.

Now we have: $C=F$, $E=F$, $F=F$, $G=F$, $H=F$, and $A=B$

From (11): $(\neg E \wedge F) \leftrightarrow (C \wedge D) = (T \wedge F) \leftrightarrow (F \wedge \downarrow F \leftrightarrow F \leftrightarrow F \vee$

Testing $A=F$, $B=F$: From (4): $(\neg A \wedge B) \rightarrow (D \wedge F) = F \rightarrow F \vee$ From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E) = F$

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Testing $A=F$, $B=F$: From (4): $(\neg A \wedge B) \rightarrow (D \wedge F) = F \rightarrow F \vee$ From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E) = F \rightarrow T \vee$ From (12): $(A \vee C) \rightarrow (B \vee \neg F) = F \rightarrow T \vee$ From (15): $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$ LHS: $T \vee F = T$ RHS: $(F \oplus D) \vee F = D \vee F = D$ So we need $D = T$

From (2): $(C \wedge D) \rightarrow \neg E = F \rightarrow T \vee$
 From (5): $(G \wedge H) \rightarrow (A \vee B) = F \rightarrow F \vee$
 From (7): $(F \oplus H) \rightarrow (D \vee E) = F \rightarrow T \vee$
 From (8): $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$
 LHS: $T \wedge T = T$
 RHS: $F \vee T = T \vee$

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RHS: $F \vee T = T \checkmark$
 From (9): $(D \wedge E) \rightarrow (\neg G \vee F) = F \rightarrow T \checkmark$
 From (13): $(G \wedge \neg H) \rightarrow (E \vee \neg A) = F \rightarrow T \checkmark$
 From (14): $(B \wedge F) \rightarrow (\neg D \vee H) = F \rightarrow F \checkmark$

Configuration 2: A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F $\checkmark$

Testing A=T, B=T: From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E) = F \rightarrow T \checkmark$ From (4): $(\neg A \wedge B) \rightarrow (D \wedge F) = F \rightarrow F \checkmark$ From (12): $(A \vee C) \rightarrow (B \vee \neg F) = T \rightarrow T \checkmark$ From (15): $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$ LHS: $F \vee F = F$ RHS: $(F \oplus D) \vee F = D$ So we need $D = F$

From (2): $(C \wedge D) \rightarrow \neg E = F \rightarrow T \checkmark$

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From (2): $(C \wedge D) \rightarrow \neg E = F \rightarrow T \checkmark$
 From (5): $(G \wedge H) \rightarrow (A \vee B) = F \rightarrow T \checkmark$
 From (7): $(F \oplus H) \rightarrow (D \vee E) = F \rightarrow F \checkmark$
 From (8): $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$
 LHS: $F \wedge F = F$
 RHS: $F \vee T = T \times$

So A=T, B=T with C=F doesn't work.

Subcase 2b: C=T, D=F From (11): $(\neg E \wedge F) \leftrightarrow F \leftrightarrow$ we need $E=T$ or $F=F$

From (1): $(A \oplus B) \rightarrow C = (A \oplus B) \rightarrow T \checkmark$ (always satisfied)

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A screenshot of a web browser window displaying a chat interface with Claude AI. The browser's address bar shows the URL 'claude.ai/chat/e11969b9-899f-49c0-b742-...'. The chat history on the left includes a puzzle titled 'Four travelers language deduction puzzle' and several logical deductions. The main chat area shows the following text: 'From (1): $(A \oplus B) \rightarrow C = (A \oplus B) \rightarrow T \checkmark$ (always satisfied)', 'Let me test $E=T, F=F$:', 'From (3): $(T \vee F) \leftrightarrow G = T \leftrightarrow G$, so $G=T$ ', 'From (10): $(H \vee T) \rightarrow (A \oplus B) = T \rightarrow (A \oplus B)$ ', 'So $A \oplus B$ must be true.', 'From (2): $(T \wedge F) \rightarrow \neg T = F \rightarrow F \checkmark$ ', and 'Testing $A=T, B=F$: From (6): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (4): $(F \wedge F) \rightarrow F = F \rightarrow F \checkmark$ From (12): $T \rightarrow (F \vee T) = T \rightarrow T \checkmark$ '. Below this, there is a partially visible line: 'Now test $A=T, B=F, C=T$: From (1): $(A \oplus B) \rightarrow C = (T \oplus F) \rightarrow T = F \rightarrow T \checkmark$ From (2): $(T \wedge F) \rightarrow \neg T = F \rightarrow F \checkmark$ From (3): $(T \vee F) \leftrightarrow G = T \leftrightarrow G \checkmark$ From (4): $(F \wedge F) \rightarrow F = F \rightarrow F \checkmark$ From (5): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (6): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (7): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (8): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (9): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (10): $(H \vee T) \rightarrow (A \oplus B) = T \rightarrow (A \oplus B) \checkmark$ From (11): $(H \vee T) \rightarrow (A \oplus B) = T \rightarrow (A \oplus B) \checkmark$ From (12): $T \rightarrow (F \vee T) = T \rightarrow T \checkmark$ '. The input field at the bottom contains the text 'How can I help you today?' and the model is identified as 'Sonnet 3.5'.

Four travelers language deduction puzzle

Testing $A=T, B=F$: From (6): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (4): $(F \wedge F) \rightarrow F = F \rightarrow F \checkmark$ From (12): $T \rightarrow (F \vee T) = T \rightarrow T \checkmark$

Now testing H: **$H=T$:** From (5): $T \rightarrow T \checkmark$ From (7): $(F \oplus T) \rightarrow T = T \rightarrow T \checkmark$ From (13): $(T \wedge F) \rightarrow T = F \rightarrow T \checkmark$ From (14): $F \rightarrow T \checkmark$ From (8): $(F \wedge T) \leftrightarrow T = F \leftrightarrow T \times$

$H=F$: From (5): $F \rightarrow T \checkmark$ From (7): $F \rightarrow T \checkmark$ From (13): $T \rightarrow T \checkmark$ From (14): $F \rightarrow T \checkmark$ From (8): $(F \wedge T) \leftrightarrow F = F \leftrightarrow F \checkmark$ From (9): $F \rightarrow T \checkmark$ From (15): $(T \vee F) \leftrightarrow (T \oplus F) \vee (T \wedge F) = T \leftrightarrow T \vee F = T \leftrightarrow T \checkmark$

Configuration 3: $A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F \checkmark$

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Configuration 3: $A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F \checkmark$

Testing $A=F, B=T$ with $C=T, D=F, E=T, F=F, G=T$: From (4): $(T \wedge T) \rightarrow (F \wedge F) = T \rightarrow F \times$

Let me also check $E=F, F=T$ with $C=T, D=F$:

From (11): $(T \wedge T) \leftrightarrow F = T \leftrightarrow F \times$

And $E=T, F=T$ with $C=T, D=F$:

From (11): $(F \wedge T) \leftrightarrow F = F \leftrightarrow F \checkmark$

From (3): $T \leftrightarrow G$, so $G=T$

From (10): $T \rightarrow (A \oplus B)$, so $A \oplus B = T$

Testing $A=T, B=F$: From (6): $T \rightarrow (T \wedge T) \checkmark$ From (12): $T \rightarrow T \checkmark$ From (4): $F \rightarrow F \checkmark$

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2. $A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F$

3. $A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F$

4. $A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F$

Step 4: Validate the argument for each configuration

Argument: $(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$

Config 1: $A=F, B=T, C=T, D=T, E=F, F=T, G=T, H=T$ LHS: $(F \vee F) = F$ RHS: $(F \vee F) \vee (T \vee F) = F \vee T = T \quad F \leftrightarrow T = \text{False} \times$

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$\vee T = T \quad T \leftrightarrow T = \text{True} \checkmark$

Config 2: $A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F$ LHS: $(T \vee F) = T$ RHS: $(F \vee F) \vee (F \vee T) = F \vee T = T \quad T \leftrightarrow T = \text{True} \checkmark$

Config 3: $A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F$ LHS: $(T \vee F) = T$ RHS: $(T \vee T) \vee (T \vee F) = T \vee T = T \quad T \leftrightarrow T = \text{True} \checkmark$

Config 4: $A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F$ LHS: $(T \vee T) = T$ RHS: $(T \vee T) \vee (T \vee F) = T \vee T = T \quad T \leftrightarrow T = \text{True} \checkmark$

Conclusion:

The valid configurations satisfying all constraints are:

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$T \vee T = T \leftrightarrow T = \text{True} \checkmark$

Conclusion:

The valid configurations satisfying all constraints are:

1. A=F, B=T, C=T, D=T, E=F, F=T, G=T, H=T
2. A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F
3. A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F
4. A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F

The argument $(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$ is:

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The argument $(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$ is:

- **FALSE** for configuration 1
- **TRUE** for configurations 2, 3, and 4

Therefore, the argument is **NOT valid** (not a tautology) as it doesn't hold for all satisfying configurations.

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I'll work through this systematically to find all valid configurations and then validate the argument.

Step 1: Analyze key constraints to narrow possibilities

Let me start by examining some critical constraint relationships.

From constraint (11): $(\neg E \wedge F) \leftrightarrow (C \wedge D)$ This means: $\neg E \wedge F$ is true iff $C \wedge D$ is true

From constraint (2): $(C \wedge D) \rightarrow \neg E$ This means: if $C \wedge D$ is true, then E is false

Combining these: If $C \wedge D$ is true, then $\neg E$ is true. For (11) to hold, we need F to be true. So: $C \wedge D \rightarrow (\neg E \wedge F)$

Also from (11): If $\neg E \wedge F$ is true, then $C \wedge D$ must be true.

Step 2: Analyze constraint (3)

$(E \vee F) \leftrightarrow G$ means:

- G is true iff at least one of E or F is true
- G is false iff both E and F are false

Step 3: Consider cases based on $C \wedge D$

Case 1: $C \wedge D = \text{True}$ From (2): $E = \text{False}$ From (11): $\neg E \wedge F$ must be true, so $F = \text{True}$ From (3): Since $F = \text{True}$, $G = \text{True}$

Now with $C=T$, $D=T$, $E=F$, $F=T$, $G=T$:

From (10): $(H \vee G) \rightarrow (A \oplus B)$ Since $G=T$, we have $T \rightarrow (A \oplus B)$, so $A \oplus B$ must be true.

From (1): $(A \oplus B) \rightarrow C$ Since $A \oplus B=T$ and $C=T$, this is satisfied.

Subcase 1a: $A=T$, $B=F$ (so $A \oplus B=T$) From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E)$ We have $T \wedge T \rightarrow T \wedge T$, which gives $T \rightarrow T \checkmark$

From (4): $(\neg A \wedge B) \rightarrow (D \wedge F)$ We have $F \wedge F \rightarrow T \wedge T$, which gives $F \rightarrow T \checkmark$

From (12): $(A \vee C) \rightarrow (B \vee \neg F)$ We have $T \rightarrow (F \vee F) = T \rightarrow F \times$

So $A=T$, $B=F$ is invalid in Case 1.

Subcase 1b: $A=F$, $B=T$ (so $A \oplus B=T$) From (4): $(\neg A \wedge B) \rightarrow (D \wedge F)$ We have $T \wedge T \rightarrow T \wedge T$, which gives $T \rightarrow T \checkmark$

From (12): $(A \vee C) \rightarrow (B \vee \neg F)$ We have $T \rightarrow (T \vee F) = T \rightarrow T \checkmark$

Now for H , let's test both values:

$H = \text{True}$: From (5): $(G \wedge H) \rightarrow (A \vee B) = T \rightarrow T \checkmark$ From (7): $(F \oplus H) \rightarrow (D \vee E) = F \rightarrow T \checkmark$ From (13): $(G \wedge \neg H) \rightarrow (E \vee \neg A) = F \rightarrow T \checkmark$ From (14): $(B \wedge F) \rightarrow (\neg D \vee H) = T \rightarrow T \checkmark$ From (15): $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$ LHS: $T \vee T = T$ RHS: $(F \oplus T) \vee (T \wedge F) = T \vee F = T \checkmark$ From (8): $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$ LHS: $T \wedge T = T$ RHS: $T \vee F = T \checkmark$ From (9): $(D \wedge E) \rightarrow (\neg G \vee F) = F \rightarrow T \checkmark$

Configuration 1: $A=F$, $B=T$, $C=T$, $D=T$, $E=F$, $F=T$, $G=T$, $H=T \checkmark$

H = False: From (5): $(G \wedge H) \rightarrow (A \vee B) = F \rightarrow T \checkmark$ From (7): $(F \oplus H) \rightarrow (D \vee E) = T \rightarrow T \checkmark$ From (13): $(G \wedge \neg H) \rightarrow (E \vee \neg A) = T \rightarrow T \checkmark$ From (14): $(B \wedge F) \rightarrow (\neg D \vee H) = T \rightarrow F \times$

So $H=F$ is invalid in Case 1.

Case 2: $C \wedge D = \text{False}$

From (11): $\neg E \wedge F$ must be false This means: $E \vee \neg F$ is true

Subcase 2a: $C=F$ From (3): Need to determine G based on E and F

From (10): $(H \vee G) \rightarrow (A \oplus B)$ If $H \vee G$ is true, then $A \oplus B$ must be true.

From (1): $(A \oplus B) \rightarrow C = (A \oplus B) \rightarrow F$ If $A \oplus B$ is true, then C must be true, but $C=F$, so $A \oplus B$ must be false.

Therefore $A \oplus B = F$, meaning $A=B$.

If $A \oplus B=F$, then from (10), $H \vee G$ must be false, so $H=F$ and $G=F$.

From (3): $(E \vee F) \leftrightarrow G = (E \vee F) \leftrightarrow F$ This means $E \vee F = F$, so $E=F$ and $F=F$.

Now we have: $C=F$, $E=F$, $F=F$, $G=F$, $H=F$, and $A=B$

From (11): $(\neg E \wedge F) \leftrightarrow (C \wedge D) = (T \wedge F) \leftrightarrow (F \wedge D) = F \leftrightarrow F \checkmark$

Testing $A=F$, $B=F$: From (4): $(\neg A \wedge B) \rightarrow (D \wedge F) = F \rightarrow F \checkmark$ From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E) = F \rightarrow T \checkmark$ From

(12): $(A \vee C) \rightarrow (B \vee \neg F) = F \rightarrow T \checkmark$ From (15): $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$ LHS: $T \vee F = T$ RHS: $(F \oplus D) \vee F = D \vee F = D$ So we need $D = T$

From (2): $(C \wedge D) \rightarrow \neg E = F \rightarrow T \checkmark$ From (5): $(G \wedge H) \rightarrow (A \vee B) = F \rightarrow F \checkmark$ From (7): $(F \oplus H) \rightarrow (D \vee E) = F \rightarrow T \checkmark$ From (8): $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$ LHS: $T \wedge T = T$ RHS: $F \vee T = T \checkmark$ From (9): $(D \wedge E) \rightarrow (\neg G \vee F) = F \rightarrow T \checkmark$ From (13): $(G \wedge \neg H) \rightarrow (E \vee \neg A) = F \rightarrow T \checkmark$ From (14): $(B \wedge F) \rightarrow (\neg D \vee H) = F \rightarrow F \checkmark$

Configuration 2: $A=F$, $B=F$, $C=F$, $D=T$, $E=F$, $F=F$, $G=F$, $H=F \checkmark$

Testing $A=T$, $B=T$: From (6): $(A \wedge \neg B) \rightarrow (C \wedge \neg E) = F \rightarrow T \checkmark$ From (4): $(\neg A \wedge B) \rightarrow (D \wedge F) = F \rightarrow F \checkmark$ From

(12): $(A \vee C) \rightarrow (B \vee \neg F) = T \rightarrow T \checkmark$ From (15): $(\neg A \vee H) \leftrightarrow (E \oplus D) \vee (G \wedge \neg C)$ LHS: $F \vee F = F$ RHS: $(F \oplus D) \vee F = D$ So we need $D = F$

From (2): $(C \wedge D) \rightarrow \neg E = F \rightarrow T \checkmark$ From (5): $(G \wedge H) \rightarrow (A \vee B) = F \rightarrow T \checkmark$ From (7): $(F \oplus H) \rightarrow (D \vee E) = F \rightarrow F \checkmark$ From (8): $(\neg A \wedge D \oplus E) \leftrightarrow (H \vee \neg F)$ LHS: $F \wedge F = F$ RHS: $F \vee T = T \times$

So $A=T$, $B=T$ with $C=F$ doesn't work.

Subcase 2b: $C=T$, $D=F$ From (11): $(\neg E \wedge F) \leftrightarrow F$, so we need $E=T$ or $F=F$

From (1): $(A \oplus B) \rightarrow C = (A \oplus B) \rightarrow T \checkmark$ (always satisfied)

Let me test $E=T$, $F=F$: From (3): $(T \vee F) \leftrightarrow G = T \leftrightarrow G$, so $G=T$

From (10): $(H \vee T) \rightarrow (A \oplus B) = T \rightarrow (A \oplus B)$ So $A \oplus B$ must be true.

From (2): $(T \wedge F) \rightarrow \neg T = F \rightarrow F \checkmark$

Testing A=T, B=F: From (6): $(T \wedge T) \rightarrow (T \wedge T) = T \rightarrow T \checkmark$ From (4): $(F \wedge F) \rightarrow F = F \rightarrow F \checkmark$ From (12): $T \rightarrow (F \vee T) = T \rightarrow T \checkmark$

Now testing H: **H=T:** From (5): $T \rightarrow T \checkmark$ From (7): $(F \oplus T) \rightarrow T = T \rightarrow T \checkmark$ From (13): $(T \wedge F) \rightarrow T = F \rightarrow T \checkmark$
From (14): $F \rightarrow T \checkmark$ From (8): $(F \wedge T) \leftrightarrow T = F \leftrightarrow T \times$

H=F: From (5): $F \rightarrow T \checkmark$ From (7): $F \rightarrow T \checkmark$ From (13): $T \rightarrow T \checkmark$ From (14): $F \rightarrow T \checkmark$ From (8): $(F \wedge T) \leftrightarrow F = F \leftrightarrow F \checkmark$ From (9): $F \rightarrow T \checkmark$ From (15): $(T \vee F) \leftrightarrow (T \oplus F) \vee (T \wedge F) = T \leftrightarrow T \vee F = T \leftrightarrow T \checkmark$

Configuration 3: A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F $\checkmark$

Testing A=F, B=T with C=T, D=F, E=T, F=F, G=T: From (4): $(T \wedge T) \rightarrow (F \wedge F) = T \rightarrow F \times$

Let me also check E=F, F=T with C=T, D=F: From (11): $(T \wedge T) \leftrightarrow F = T \leftrightarrow F \times$

And E=T, F=T with C=T, D=F: From (11): $(F \wedge T) \leftrightarrow F = F \leftrightarrow F \checkmark$ From (3): $T \leftrightarrow G$, so $G=T$

From (10): $T \rightarrow (A \oplus B)$, so $A \oplus B=T$

Testing A=T, B=F: From (6): $T \rightarrow (T \wedge T) \checkmark$ From (12): $T \rightarrow T \checkmark$ From (4): $F \rightarrow T \checkmark$

H=T: From (5): $T \rightarrow T \checkmark$ From (7): $F \rightarrow T \checkmark$ From (13): $F \rightarrow T \checkmark$ From (14): $F \rightarrow T \checkmark$ From (8): $(F \wedge T) \leftrightarrow T = F \leftrightarrow T \times$

H=F: From (8): $F \leftrightarrow F \checkmark$ From (9): $F \rightarrow T \checkmark$ From (15): $T \leftrightarrow T \vee F = T \leftrightarrow T \checkmark$

Configuration 4: A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F $\checkmark$

Testing A=F, B=T with C=T, D=F, E=T, F=T, G=T: From (4): $T \rightarrow F \times$

And E=F, F=F with C=T, D=F: From (3): $F \leftrightarrow G$, so $G=F$ From (10): $(H \vee F) \rightarrow (A \oplus B) = H \rightarrow (A \oplus B)$

If H=F: Then $A \oplus B$ can be anything From (11): $T \leftrightarrow F \times$

If H=T: $A \oplus B$ must be true From (11): $T \leftrightarrow F \times$

So this doesn't work.

Summary of Valid Configurations:

1. A=F, B=T, C=T, D=T, E=F, F=T, G=T, H=T
2. A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F
3. A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F
4. A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F

Step 4: Validate the argument for each configuration

Argument: $(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$

Config 1: A=F, B=T, C=T, D=T, E=F, F=T, G=T, H=T LHS: $(F \vee F) = F$ RHS: $(F \vee F) \vee (T \vee F) = F \vee T = T$ $F \leftrightarrow T =$

False $\times$

Config 2: A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F LHS: $(T \vee F) = T$ RHS: $(F \vee F) \vee (F \vee T) = F \vee T = T \leftrightarrow T =$
True ✓

Config 3: A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F LHS: $(T \vee F) = T$ RHS: $(T \vee T) \vee (T \vee F) = T \vee T = T \leftrightarrow T =$
True ✓

Config 4: A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F LHS: $(T \vee T) = T$ RHS: $(T \vee T) \vee (T \vee F) = T \vee T = T \leftrightarrow T =$
True ✓

Conclusion:

The valid configurations satisfying all constraints are:

1. A=F, B=T, C=T, D=T, E=F, F=T, G=T, H=T
2. A=F, B=F, C=F, D=T, E=F, F=F, G=F, H=F
3. A=T, B=F, C=T, D=F, E=T, F=F, G=T, H=F
4. A=T, B=F, C=T, D=F, E=T, F=T, G=T, H=F

The argument $(\neg B \vee F \oplus H) \leftrightarrow (\neg D \vee E) \vee (G \vee \neg C)$ is:

- **FALSE** for configuration 1
- **TRUE** for configurations 2, 3, and 4

Therefore, the argument is **NOT valid** (not a tautology) as it doesn't hold for all satisfying configurations.